

DEEPAK ARORA

Principal Systems & Platform Architect • Linux Kernel & Network Datapath • GPU/AI Compute

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SUMMARY

Principal systems architect with 20+ years owning the platform substrate that large fleets run on — Linux kernel and network datapath, GPU and AI compute, and security architecture — from silicon-level decisions through multi-year build-vs-buy strategy. Led 25+ cross-functional HW/FW/SW teams at Verizon delivering across a 4M+ device fleet, root-causing systemic field failures back into architecture rather than point fixes. Now owns end-to-end architecture for a greenfield AI-native platform taken from zero to production in five months, with cost per unit of work engineered into the datapath as a first-class design constraint.

SCOPE & AUTHORITY

- **Teams led** — 25+ cross-functional HW/FW/SW engineering teams at Verizon; currently leads a 10-person engineering team and a product manager from architecture through launch.
- **Production scale** — architecture decisions running across 3M+ Fios Video and 1M+ OTT devices; GPU pipelines processing 2,000+ hours of audio; a marketplace past 1M lifetime orders.
- **Decision authority** — GPU hardware and provider selection, build-vs-buy and TCO, datapath and offload architecture, security architecture, multi-year technology roadmap.
- **Artifacts owned** — production architecture specifications ODM and internal teams build against, benchmark-driven provider evaluations, capacity and cost models, architecture review boards, whitepapers.
- **Intellectual property** — one granted US patent; one published whitepaper with a provisional application filed.

PROFESSIONAL EXPERIENCE

Senior AI Systems Architect | SwifTrade — Irving, TX

Feb 2026 – Present

Platform architecture owner for a greenfield AI-native B2B marketplace; leads a 10-person engineering team and a product manager.

PLATFORM & GPU INFRASTRUCTURE

- Architected and launched SwifTradeSupply.com end to end — system decomposition, data model, integration and security architecture, launch sequencing — taking it from zero to production in five months; shipped July 2026.
- Selected the GPU and AI compute substrate on benchmarked stability, cost per GPU-hour, model support and API-driven provisioning; that decision now governs every downstream AI system on the platform.
- Engineered cost per unit of work directly into the datapath: cheapest-first source layering that reaches paid APIs only after free sources are exhausted, response caching with in-flight request coalescing, and per-run unit-cost accounting.

AI SYSTEMS IN PRODUCTION

- Built GPU-accelerated catalog automation — multi-storefront listing generation via lexical and semantic matching with GPU image similarity, plus master-category classification using embeddings and clustering — cutting supplier onboarding from days to minutes.
- Architected a multi-agent buyer-acquisition pipeline on a LangGraph state machine — geo-targeted discovery, multi-source contact enrichment, explainable lead scoring, personalized outreach — instrumented with LangSmith tracing for run-level observability and failure triage; validated on a 1,379-business pilot and now in controlled live operation.
- Designed the output-governance layer for generated outbound: constrained assembly keeping every claim traceable to a verified source, deterministic pre-send validation with a human review queue, a readiness gate that halts unsafe configuration, and automated reply classification driving suppression.
- Scaled Amazon-channel fulfillment 6.6× (300 → 2,000 orders/day) on the automation layer built in-house, delivering ~300K incremental orders over the following six months against a platform lifetime total past 1M.

ENGINEERING LEADERSHIP

- Led and developed a 10-person engineering team, establishing architecture review practice, AI-assisted development standards and QA automation that measurably accelerated production delivery.
- Built a company-wide model-benchmark framework scoring coding models and agentic harnesses on cost, code correctness, iterations to convergence and latency, then standardized the toolchain and an automated code-review loop on the results.

Principal Engineer — Embedded Systems & AI Workflows | Verizon — Irving, TX

2017 – Dec 2025

Architecture lead for consumer video and edge platforms; 3M+ Fios Video and 1M+ OTT devices in the field.

GPU INFRASTRUCTURE & AI PIPELINES

- Architected a Whisper ASR and LLM analysis pipeline on NVIDIA Tesla GPUs for early field-issue detection, processing 2,000+ hours of noisy narrowband telephony at 75% transcription accuracy with CUDA workloads tuned for throughput and cost per hour of audio; authored the business case quantifying a ~25% call-deflection opportunity.
- Ran the comparative datapath analysis for high-rate packet ingestion into GPU-accelerated pipelines — DPDK kernel-bypass versus SmartNIC/DPU offload — scored against latency budgets, throughput ceilings and per-node hardware cost; the result set the platform's offload

direction.

- Diagnosed DGX A100 (8×A100, 640 GB) multi-GPU server and ProLiant DL385 subsystem failures via BMC/iLO and IPMI, hardening the GPU compute tier.

PLATFORM ARCHITECTURE & TECHNICAL LEADERSHIP

- Led 25+ cross-functional HW/FW/SW engineering teams delivering quarterly releases across Stream TV, set-top boxes, soundbars, remote devices and CE applications; developed engineers through architecture review.
- Authored the production-grade architecture and design specifications for thin-client hardware, MEC deployment, security and A/V pipelines — the reference documents ODM and internal teams built against.
- Architected an edge-native virtual mobile computing platform (independent R&D): thin-client hardware, Android containerization at the MEC tier, latency-sensitive A/V transport and zero-trust security; advanced through prototyping, published as a whitepaper, provisional patent application filed.
- Owned regulatory and industry certification programs end to end — FCC Part 15 (B/C/E), UL 62368-1, Wi-Fi Alliance, Bluetooth SIG, Energy Star — from design review through lab submission and grant.

PLATFORM RELIABILITY & NON-FUNCTIONAL REQUIREMENTS

- Root-caused systemic field failures across a 4M+ device fleet — MoCA link instability, 802.11 interference, A/V dropouts, DVR faults, update freezes — driving causes back into platform architecture rather than point fixes.
- Designed the DVR high-throughput storage datapath sustaining 24 parallel I/O streams under heavy disk contention, with fragmentation mitigation and write-coalescence algorithms for real-time record and playback.

Principal Engineer — Embedded Systems | Verizon — San Jose, CA

2014 – 2017

- Designed eMMC flash layouts, boot state machines and update frameworks, cutting field update time by ~15 seconds.
- Built bootloaders, device drivers, IR firmware and platform diagnostics that measurably improved field reliability.

Senior Embedded Platform Engineer (Contract) | Intel Corporation — Portland, OR

2010 – 2014

- Engineered Linux kernel modules and high-performance middleware tuned for memory, IPC, networking and real-time constraints.
- Executed ARM/x86 board bring-up, MoCA 2.0 PHY/MAC driver debugging and eMMC flash architecture.

EARLIER EXPERIENCE

- **Technical Lead, HCL Technologies** (2008 – 2010) — Raised L3 routing throughput using Cavium Octeon hardware acceleration for packet classification and forwarding offload; built an enterprise Windows security platform for 500–1,000 network endpoints.
- **Senior Software Engineer, GlobalLogic** (2007 – 2008) — Built VoIP/SBC middleware in C/C++; debugged DTMF and implemented STB DNS caching.
- **Software Engineer, Conexant Systems** (2005 – 2007) — Ported full TCP/IP and IPv6 (RFC 2461 Neighbor Discovery) stacks to Nucleus/Quantum RTOS; improved 802.11 PHY/MAC performance.

CORE TECHNICAL SKILLS

- **AI & GPU Compute:** PyTorch, CUDA optimization and profiling, ASR (Whisper) and LLM inference pipelines, RAG, embeddings and clustering, multi-agent pipelines (LangGraph) with LangSmith tracing, model benchmarking, AI governance (traceability, bias testing); DGX A100, NVIDIA Tesla, managed GPU providers
- **Kernel, Datapath & Networking:** Linux kernel modules, network and device drivers, DPDK kernel-bypass, SmartNIC/DPU offload, hardware packet-processing offload (Cavium Octeon), TCP/IP and IPv6 (RFC 2461), L3 routing and NAT, packet classification, DNS caching, 802.3 Ethernet, 802.11 Wi-Fi PHY/MAC, MoCA 2.0, VoIP/SBC, TR-069/TR-369, PCIe, real-time and latency-budget tuning
- **Security & Compliance:** Zero-trust architecture, secure boot and chain of trust, ARM TEE, firmware hardening, Windows endpoint security, SAST/DAST, penetration testing, Black Duck SCA; FCC Part 15 (B/C/E), UL 62368-1, Wi-Fi Alliance, Bluetooth SIG, RoHS, Energy Star, CPSC
- **Platforms, Cloud & Tooling:** C, C++, Python, Bash, ARM/x86 assembly; Linux kernel, Yocto/BitBake, U-Boot and bootloaders, BSPs and board bring-up, OpenBMC, BIOS, Android/HAL, RTOS (Nucleus, Quantum); AWS, GCP, Azure, Docker, Kubernetes; Git, Gerrit, Jenkins CI, GDB, Valgrind, Wireshark, JTAG/UART, BMC/iLO, IPMI
- **Architecture & Leadership:** Reference architectures and design specifications, architecture review boards, build-vs-buy and TCO analysis, technology roadmapping, vendor and ODM management, cross-functional team leadership, engineer development

PATENTS, PUBLICATIONS & OPEN SOURCE

- **Granted patent** — US 10,606,605 B2: Software Upgrade & Disaster Recovery (2020).
- **Whitepaper / provisional** — Virtual Mobile Computing via Ultra-High-Speed, Low-Latency 5G MEC (whitepaper 2022; provisional application filed 2026).
- **Open source** — github.com/deepakaroraembedded-design — [noun-rank](#) [cross-model summary validation](#) · [stock-market analysis](#) · [ResumeRanker](#).

EDUCATION

MCA, Master of Computer Applications — GGSIP University (2004) · **B.Sc. Mathematics** — Hansraj College, Delhi University (2001)

Quantitative Modeling (UPenn, 2025) · GenAI Ready (Verizon, 2024) · Strategic Thinking (Harvard, 2021) · Business Case Development (Harvard, 2021)